

Math 162

Problem set 8 solutions

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1. (a) By Problem Set 5, #2(a) we have

$$\lim_{n \rightarrow \infty} \frac{\log n}{n} = \lim_{x \rightarrow \infty} \frac{\log x}{x} = 0.$$

Moreover, we have

$$\frac{d}{dx} \frac{\log x}{x} = \frac{1 - \log x}{x^2},$$

from which it follows that the sequence $(\frac{\log n}{n})$ is decreasing for $n \geq 3$. So the Leibniz test implies that the series converges.

- (b) We compute

$$\lim_{n \rightarrow \infty} \frac{\sin 1/n}{1/n} = \lim_{x \rightarrow \infty} \frac{\sin 1/x}{1/x} = \lim_{x \rightarrow 0^+} \frac{\sin x}{x} = \lim_{x \rightarrow 0^+} \frac{\cos x}{1} = 1,$$

where we used L'Hôpital's rule for the second-to-last equality. Note that $\sin \frac{1}{n} > 0$ for all $n \in \mathbb{N}$, so the limit comparison test applies. Since the harmonic series diverges, the series in question diverges.

- (c) Since $\frac{1}{x(\log x)^2} > 0$ for $x > 1$, we can apply the integral test. We compute

$$\int_2^\infty \frac{dx}{x(\log x)^2} = \int_{\log 2}^\infty \frac{du}{u^2} = -\frac{1}{u} \Big|_{\log 2}^\infty = \frac{1}{\log 2}.$$

Since this improper integral converges, so does the series.

- (d) Observe that $0 < \frac{1}{n^2 \log n} < \frac{1}{n^2}$ for all $n \geq 2$, so our series converges by the comparison test.
- (e) We apply the ratio test. Observe that

$$\lim_{n \rightarrow \infty} \frac{(n+1)!/(n+1)^{n+1}}{n!/n^n} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n.$$

By Problem Set 5, #2(c), we have

$$\lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = e,$$

which implies that

$$\lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} \frac{1}{((n+1)/n)^n} = \frac{1}{e} < 1.$$

Thus the ratio test implies that our series converges.

2. For any $1 \leq k \leq n$, we have $1 \leq n - k + 1 \leq n$, and hence $1 \leq \sqrt{k}, \sqrt{n - k + 1} \leq \sqrt{n}$. Thus

$$\frac{1}{\sqrt{k}} \cdot \frac{1}{\sqrt{n - k + 1}} \geq \frac{1}{\sqrt{n}} \cdot \frac{1}{\sqrt{n}} = \frac{1}{n},$$

which allows us to estimate

$$|c_n| = \left| \sum_{k=1}^n (-1)^{n+1} \frac{1}{\sqrt{k}} \cdot \frac{1}{\sqrt{n-k+1}} \right| = \sum_{k=1}^n \frac{1}{\sqrt{k}} \cdot \frac{1}{\sqrt{n-k+1}} \geq \sum_{k=1}^n \frac{1}{n} = 1.$$

It follows that the sequence (c_n) does not tend to 0, so the corresponding series diverges.

3. Our hypothesis implies that $\lim_{n \rightarrow \infty} \frac{a_n}{1+a_n} = 0$. In particular, there exists $N \in \mathbb{N}$ such that $n > N$ implies that $\frac{a_n}{1+a_n} < \frac{1}{2}$. Since $a_n \geq 0$, we have $1 + a_n \geq 1 > 0$, which allows us to rearrange the previous inequality to obtain $a_n < 1$ for $n > N$. But then we have $\frac{a_n}{1+a_n} > \frac{a_n}{2}$ and hence $a_n < 2\frac{a_n}{1+a_n}$ for $n > N$, so $\sum_{n=1}^{\infty} a_n$ converges by the comparison test.

4. (a) We simply compute that

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{p_n}{p_{n-1}} = \frac{\lim_{n \rightarrow \infty} p_n}{\lim_{n \rightarrow \infty} p_{n-1}} = 1.$$

Here the second equality is justified because $\lim_{n \rightarrow \infty} p_n \neq 0$.

- (b) Observe that

$$\log p_n = \sum_{k=1}^n \log a_k.$$

If $\lim_{n \rightarrow \infty} p_n$ exists and is nonzero (hence positive), then

$$\lim_{n \rightarrow \infty} \log p_n = \log \left(\lim_{n \rightarrow \infty} p_n \right)$$

by the continuity of $\log$. Thus $\sum_{n=1}^{\infty} \log a_n$ converges. Conversely, if the latter series converges, then we have

$$\lim_{n \rightarrow \infty} \log p_n = \sum_{n=1}^{\infty} \log a_n$$

and hence

$$\lim_{n \rightarrow \infty} p_n = \exp \left(\lim_{n \rightarrow \infty} \log p_n \right) = \exp \left(\sum_{n=1}^{\infty} \log a_n \right)$$

by the continuity of $\exp$.

- (c) Assume first that the infinite product converges. Since $b_n \geq 0$ for any $n \in \mathbb{N}$, it suffices to show that the sequence of partial sums $s_n = \sum_{k=1}^n b_k$ is bounded. Note that the partial product can be expanded as

$$p_n = \prod_{k=1}^n (1 + b_k) = 1 + \sum_{k=1}^n b_k + \cdots,$$

where the remaining terms are products of two or more factors b_k . Thus we have $s_n \leq p_n$ for all $n \in \mathbb{N}$. Since the sequence (p_n) converges, it is bounded, and hence (s_n) is bounded.

Now suppose that the series converges. Note that for all $x \geq 0$, we have $\log(1+x) \leq x$. For instance, this follows from the definition of $\log$ as an integral:

$$\log(1+x) = \int_1^{1+x} \frac{dt}{t} \leq \int_1^{1+x} dt = x.$$

Thus we have $0 \leq \log(1+b_n) \leq b_n$ for all $n \in \mathbb{N}$, so the comparison test implies that the series $\sum_{n=1}^{\infty} \log(1+b_n)$ converges. Apply part (b) to finish the proof.

5. We give an example where f is continuous; this also answers part (a). Define $f : [0, \infty) \rightarrow \mathbb{R}$ by

$$f(x) := \begin{cases} n^2x + 1 - n^3 & \text{for } n - \frac{1}{n^2} \leq x \leq n, \ n \geq 2, \\ -n^2x + 1 + n^3 & \text{for } n \leq x \leq n + \frac{1}{n^2}, \ n \geq 2, \\ 0 & \text{otherwise.} \end{cases}$$

Thus f is piecewise linear and continuous, with graph consisting of an isosceles triangle with base $\frac{2}{n^2}$ and height 1 centered at n for each $n \geq 2$. Since each such triangle has area $\frac{1}{n^2}$, it follows that

$$\int_0^\infty f(x) \, dx = \sum_{n=2}^\infty \frac{1}{n^2}$$

converges. But certainly $\lim_{x \rightarrow \infty} f(x)$ does not exist, since there exist arbitrarily large values of x with $f(x) = 0$ or $f(x) = 1$.